

Deep AWS Concepts (Interview-Level Understanding)

1. What is Amazon Web Services?

AWS is a **cloud computing platform** that allows you to use IT resources (servers, storage, databases, networking) over the internet instead of managing physical hardware.

 Traditional way:

- Buy server
- Maintain it
- Scale manually

 AWS way:

- Rent server instantly
- Scale automatically
- Pay only for what you use

 Real-world understanding:

If you build an app:

- Instead of buying a server → use AWS EC2
- Instead of hard disk → use S3
- Instead of managing DB → use RDS

 AWS = **data center on rent**

2. What is Amazon EC2?

EC2 is a **virtual machine (server)** in the cloud.

 You can:

- Choose OS (Linux/Windows)
- Install software

- Run applications

💡 Real-world example:

You built a backend (Spring Boot):

- You deploy it on EC2
- Users access it via public IP

🔥 Key concepts:

- Instance = virtual server
- Instance type = CPU + RAM configuration
- AMI = pre-configured OS image

👉 Think:

EC2 = “your laptop/server in cloud”

🟢 3. What is Amazon S3?

S3 is an **object storage service** used to store files.

👉 Stores:

- Images
- Videos
- Documents
- Backups

💡 Real-world example:

In your app:

- User uploads profile photo → stored in S3
- Backend stores only URL

🔥 Key features:

- Unlimited storage
- High durability (almost never lose data)

- Accessible via URL

👉 Think:

S3 = “Google Drive for your application”

● 4. EC2 vs S3 (Conceptual Difference)

EC2

S3

Compute (run app) Storage (store files)

Has OS

No OS

You manage server Fully managed

👉 Simple:

- EC2 → runs your backend
 - S3 → stores your files
-

● 5. What is AWS Identity and Access Management (IAM)?

IAM is the **security system of AWS**.

👉 It controls:

- Who can access AWS
- What they can do

💡 Example:

- Developer → full access
- Tester → read-only
- EC2 → access S3

🔥 Key concepts:

- User → person
- Role → permission for service
- Policy → rules

👉 Think:

IAM = “login + permissions system”

🟢 6. What is Region & Availability Zone?

AWS is divided into:

🌐 **Region:**

- Physical location (e.g., Mumbai)

🏢 **Availability Zone (AZ):**

- Data centers inside region

💡 **Example:**

Mumbai region → multiple AZs

👉 Why important?

If one AZ fails → app still works

👉 Think:

- Region = city
 - AZ = buildings inside city
-

🟢 7. What is Amazon VPC?

VPC is your **private network inside AWS**.

👉 You control:

- IP addresses
- Subnets
- Security

💡 **Real-world:**

Like your home network:

- Private space

- Controlled access

👉 Used to:

- Secure your servers
 - Separate public & private resources
-

🟢 8. What is Subnet?

Subnet = division of VPC.

👉 Types:

- Public subnet → internet access
- Private subnet → no direct internet

💡 Example:

- Web server → public subnet
- Database → private subnet

👉 Security design 🔥

🟢 9. What is Load Balancer?

Load Balancer distributes traffic across multiple servers.

👉 Problem:

1 server → overload → crash

👉 Solution:

Load Balancer:

- Divides traffic
- Improves performance

👉 Think:

Like traffic police managing vehicles

● 10. What is Auto Scaling?

Auto Scaling automatically:

- Adds servers when load increases
- Removes servers when load decreases

💡 Example:

Flipkart sale:

- Traffic increases → more servers

👉 Saves cost + improves performance

● 11. What is Amazon RDS?

RDS is a **managed database service**.

👉 AWS handles:

- Setup
- Backup
- Maintenance

💡 Example:

Instead of installing MySQL manually → use RDS

👉 Think:

RDS = “Database without headache”

● 12. What is AWS Lambda?

Lambda lets you run code **without managing servers**.

👉 You just:

- Write function
- AWS runs it

💡 Example:

- Upload image → Lambda processes it

👉 Pay only when code runs

● 13. What is High Availability?

System remains available even if one component fails.

👉 Achieved using:

- Multiple AZs
 - Load Balancer
-

● 14. What is Fault Tolerance?

System continues working even after failure.

👉 Example:

One server down → others still serve

● 15. What is Scalability?

Ability to handle increasing users.

👉 Types:

- Vertical → increase power
 - Horizontal → increase servers
-

🔥 MOST IMPORTANT (INTERVIEW GOLD)

● 16. Explain Real System Design (VERY IMPORTANT)

👉 “How would you deploy your app on AWS?”

Answer:

- Frontend → S3

- Backend → EC2
- Database → RDS
- Files → S3
- Security → IAM
- Traffic → Load Balancer
- Scaling → Auto Scaling

👉 This answer impresses interviewer **100**

🎯 Final Advice

👉 Don't mug up services

👉 Understand **how everything connects**

🚀 Next Step

If you want:

👉 I'll give **scenario-based AWS questions** (real interview level)

👉 Or take your **mock interview** (React + AWS + Backend)

Just tell me 👍

give more only 16 are not sufficient

Perfect 👍 — now you're thinking in the **right direction** (deep understanding).
Let's continue with **MORE AWS concepts** (interview-level + real understanding)
beyond the first 16.

📖 Advanced AWS Concepts (17–35) – Deep Understanding

🟡 17. What is Amazon EBS?

EBS (Elastic Block Store) is **block storage attached to EC2**, like a hard disk.

👉 Key idea:

- EC2 = computer
- EBS = hard disk

💡 Real-world:

When you install software on EC2 → it is stored in EBS

🔥 Important:

- Data persists even if EC2 stops
- Can take snapshots (backup)

🟡 18. EBS vs S3 (Important)

EBS

S3

Attached to EC2 Independent storage

Block storage Object storage

Fast, low latency Scalable storage

👉 Use:

- EBS → OS, application files
- S3 → images, videos

🟡 19. What is AMI (Amazon Machine Image)?

AMI is a **pre-configured template** used to launch EC2.

👉 Contains:

- OS (Linux/Windows)
- Software
- Configuration

💡 Example:

You install Java + app → create AMI → reuse later

● 20. What is Security Group?

Security Group acts as a **firewall** for EC2.

👉 Controls:

- Incoming traffic (inbound)
- Outgoing traffic (outbound)

💡 Example:

Allow:

- Port 22 → SSH
- Port 80 → HTTP

👉 Important:

- It is **stateful** (automatically allows response)

● 21. What is NACL (Network ACL)?

NACL is a **firewall at subnet level**.

👉 Difference:

- Security Group → instance level
- NACL → subnet level

👉 Important:

- Stateless (need rules for both directions)

● 22. Security Group vs NACL (Concept)

Feature Security Group NACL

Level	Instance	Subnet
Type	Stateful	Stateless

👉 Interview tip:
Mention both differences + use case

● 23. What is Internet Gateway?

Internet Gateway allows your VPC to **connect to the internet**.

👉 Without it:

- No public access

💡 **Example:**

EC2 with public IP needs IGW to access internet

● 24. What is NAT Gateway?

NAT Gateway allows **private subnet instances to access internet**, but prevents incoming traffic.

👉 Example:

- Private DB server → download updates → via NAT

👉 Security benefit 🔥

● 25. What is Route Table?

Route table defines **how traffic moves inside VPC**.

👉 Example:

- Internet traffic → Internet Gateway
 - Internal traffic → private subnet
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● 26. What is Amazon CloudWatch?

CloudWatch is used for **monitoring AWS resources**.

👉 Tracks:

- CPU usage
- Memory
- Logs

💡 **Example:**

If CPU > 80% → trigger alert

● **27. What is Logging in AWS?**

Logging helps track system activity.

👉 **Tools:**

- CloudWatch logs
- S3 logs

👉 **Used for:**

- Debugging
 - Security
-

● **28. What is Amazon CloudFront?**

CloudFront is a **Content Delivery Network (CDN)**.

👉 It delivers content from nearest location to user.

💡 **Example:**

User in India → content served from India, not US

👉 **Result:**

- Faster loading
-

● **29. What is Amazon API Gateway?**

API Gateway is used to:

- Create APIs

- Manage API traffic
- Secure APIs

👉 Works with Lambda or backend

● 30. What is Serverless Architecture?

Serverless means:

- No server management
- AWS handles everything

👉 Example:

Lambda + API Gateway

👉 Benefit:

- Pay only when used
-

● 31. What is Amazon SQS?

SQS is a **message queue service** used to decouple systems.

💡 Example:

- Order placed → goes to queue
- Backend processes later

👉 Prevents system overload

● 32. What is Amazon SNS?

SNS is used for **sending notifications**.

👉 Example:

- Email
- SMS
- Push notifications

● 33. What is Decoupling?

Decoupling means:

👉 Different components work independently

👉 Example:

Frontend → Queue → Backend

👉 Benefits:

- Better scalability
- Fault tolerance

● 34. What is Disaster Recovery?

Disaster Recovery ensures system can recover after failure.

👉 Techniques:

- Backup (snapshots)
- Multi-region deployment

● 35. What is Multi-AZ vs Multi-Region?

Multi-AZ

Multi-Region

Within same region Across regions

High availability Disaster recovery

🔥 **BONUS (VERY IMPORTANT)**

● 36. Explain Real Application Flow (Advanced)

👉 Interview Question:

“How your app works on AWS?”

Answer:

1. User opens React app → hosted on S3
2. Request goes to backend → EC2
3. Data stored in → RDS
4. Files stored in → S3
5. Load handled by → Load Balancer
6. Scaling via → Auto Scaling
7. Monitoring → CloudWatch

👉 This is **gold answer** 100